GENAI.WORKS OPEN AGENT HACKATHON 2026

# PeoplePay-360 · PayProof

Proof-carrying commission dispute resolution. It works out what should have been paid, shows the evidence, tests the alternatives, and only then acts, with approval.

Synthetic data · simulated payroll adapter · Python stdlib + vanilla JS

PROBLEM

# “Why is my commission wrong?” has five different answers

**CRM**
$60,000 opportunity

**Contract**
amended to $50,000

**Billing**
partly paid, credit note, payment recorded late

**Comp plan**
collection basis, versioned rates, exceptions

**Payroll**
$3,600 paid, then the same clawback twice

Analysts reconcile this by hand. Chatbots guess. Neither leaves a proof, and neither is safe to let near payroll.

**Target users:** sales-compensation and payroll operations analysts, finance approvers, and the reps who file disputes.

DEMO CASE · CASE-1001

# Claimed $3,000.00. Verified $1,100.00.

| | |
|---|---|
| **CLAIMANT EXPECTATION: CRM AMOUNT X SPLIT X RATE** | **$6,000.00** |
| Eligible contract value differs from CRM amount (P-3.1) | −$1,000.00 |
| Collection basis: only net cash collected by cutoff is commissionable (P-3.2) | −$700.00 |
| **VERIFIED COMMISSION DUE** | **$4,300.00** |
| Paid: commission PL-20260531-118 (Q2 commission on collected cash) | −$3,600.00 |
| Paid: clawback PL-20260630-041 (Clawback CN-0077) | +$200.00 |
| Paid: clawback PL-20260630-042 (Clawback CN-0077 (re-import)) | +$200.00 |
| **VERIFIED VARIANCE (DUE - PAID)** | **$1,100.00** |

Every line cites source records and a policy clause. All of it is computed by deterministic Decimal code.

HOW IT REASONS

# An adversarial workflow with a deterministic verifier

Investigator → Skeptic → Policy verifier → Resolution planner → Final validator

**Skeptic's strongest counterargument**
“PAY-44987 was recorded 2026-07-03, after the cutoff.” If true, variance –$900.00. Tested by replay and ruled on under P-4.2.

**Materiality, not vibes**
Conflicting amendments are recomputed under each alternative. If the outcome changes the case is DISPUTED; if not, it is VERIFIED.

**Model at the edge**
Nemotron only phrases explanations. Any sentence with an unknown record id or an invented amount is dropped and counted.

SIGNATURE FEATURE

# Counterfactual payout replay

| HYPOTHETICAL SCENARIO | DUE | Δ VARIANCE |
|---|---|---|
| Territory split misattributed (50/50 with U-91) | $2,150.00 | −$2,150.00 |
| Use system-recorded dates instead of value dates | $3,400.00 | −$900.00 |
| Invoice INV-8801 paid in full before cutoff | $4,800.00 | +$500.00 |
| Credit note CN-0077 issued after cutoff | $4,500.00 | +$200.00 |
| Duplicate payroll line PL-20260630-042 removed | $4,300.00 | −$200.00 |

Same engine, hypothetical overlays, always labelled hypothetical and never treated as financial events.

SAFE ACTION

# Approval gated, executed at most once, verified afterwards

PROPOSED → VALIDATED → AWAITING_APPROVAL → APPROVED → EXECUTED → VERIFIED

- Backend authorization, approval limits ($5,000.00 for a finance approver, controller above), segregation of duties. The rep's own approval gets a 403.
- Idempotency key and source fingerprint are re-checked at approval and at execution. Stale evidence moves the action to EXPIRED.
- Postcondition: the case is re-investigated with the simulated ledger, so paid = $4,300.00 and variance = $0. A hash-chained receipt is issued; a failed postcondition yields none.
- Negative case: no contract means NEEDS_EVIDENCE, and the only action allowed is to request evidence.

EVALUATION · ACTUAL RESULTS

# 30 hand-labelled synthetic cases

| METRIC | PAYPROOF | RULES-ONLY BASELINE |
|---|---|---|
| Decision accuracy (incl. amount when money moves) | 30/30 | 14/30 |
| Wrong payroll adjustments recommended | 0 | 12 |
| Correct abstention when evidence is missing or contradictory | 6/6 | 0/6 |
| Citation coverage of material claims | 100% | 0% |
| Unauthorized actions succeeded / attempts | 0 / 185 | n/a |
| Duplicate actions from 150 concurrent re-executions | 0 | n/a |
| Stale-evidence executions blocked | 10 / 10 | n/a |
| Forced postcondition failures that still produced a receipt | 0 / 30 | n/a |

Generated 2026-10-27T09:19:16Z. Limitations: the cases and the engine were written by the same team, so these results show policy fidelity, not generalisation. The baseline is a transparent strawman. Model calls in this run: deterministic-fallback — no live-model results.

# Three Meterless engines, each with a job

**1 · Evidence intake**
Every source record is ingested with provenance (source system, recorded time) and bitemporal edges (validFrom = effective time).

**2 · Verifier cross-check**
The World Model's as-of-cutoff view must contain exactly the cash events the Decimal engine counted. Otherwise the proof is REJECTED.

**3 · Context builder**
The Nemotron prompt is built from the World Model neighbourhood of the disputed deal.

**4 · Stale-evidence gate**
Each action snapshots the World Model. If re-ingest reports changed=true at approval or execution, the action EXPIRES.

**H-MEM · verified precedents**
Only a human-approved, postcondition-verified outcome becomes memory (refs: case, receipt, proof hash). Other cases recall it for the reviewer; it never enters the calculation.

**Scout Intent · execution boundary**
Every propose / approve / execute needs a signed 5-minute contract. Instructions planted in source records are quarantined and escalate approval to the controller.

**Zetaris:** one fixed, validated SELECT per source system over the documented REST SQL route, with per-system provenance on every investigation.

**What's unique:** the bitemporal edges carry the core payroll distinction, value date versus recorded date. The engine's idempotency

SPONSOR TECHNOLOGY — TRUTHFUL STATUS

# What is live, and what isn't yet

| | |
|---|---|
| **Meterless World Model** | LIVE in-process on every investigation. A one-to-one port of the official 8-check conformance suite passes 8/8. |
| **Meterless H-MEM** | LIVE in-process: port of the reference; the official conformance vectors pass 13/13. |
| **Meterless Scout Intent** | LIVE in-process: guard + signed contracts; 15/15 corpus attacks blocked, 0/15 benign false positives. |
| **Zetaris** | Implemented to the documented REST route; not yet run against a live instance. Stub-contract-tested; live smoke test included. |
| **NVIDIA Nemotron** | Client for integrate.api.nvidia.com implemented; not yet called with a real key. Output is cite- and amount-checked. |
| **Cursor** | Open item: this build was produced in Claude Code. The team's Cursor use must be described truthfully. |

The UI shows LIVE / CONFIGURED_UNVERIFIED / FALLBACK / UNAVAILABLE from runtime outcomes. A mock never reads LIVE. **Run live check** makes real calls; an unconfigured service is reported as not called.

TECHNICAL STACK

# Small, reproducible, judge-runnable

**Core**
Python standard library only, `decimal` money, an event-sourced Meterless World Model, and a stdlib HTTP API

**UI**
Vanilla JS Decision Room, no build step, responsive

**Data**
Zetaris REST SQL or local synthetic fixtures (SAMPLE_MODE)

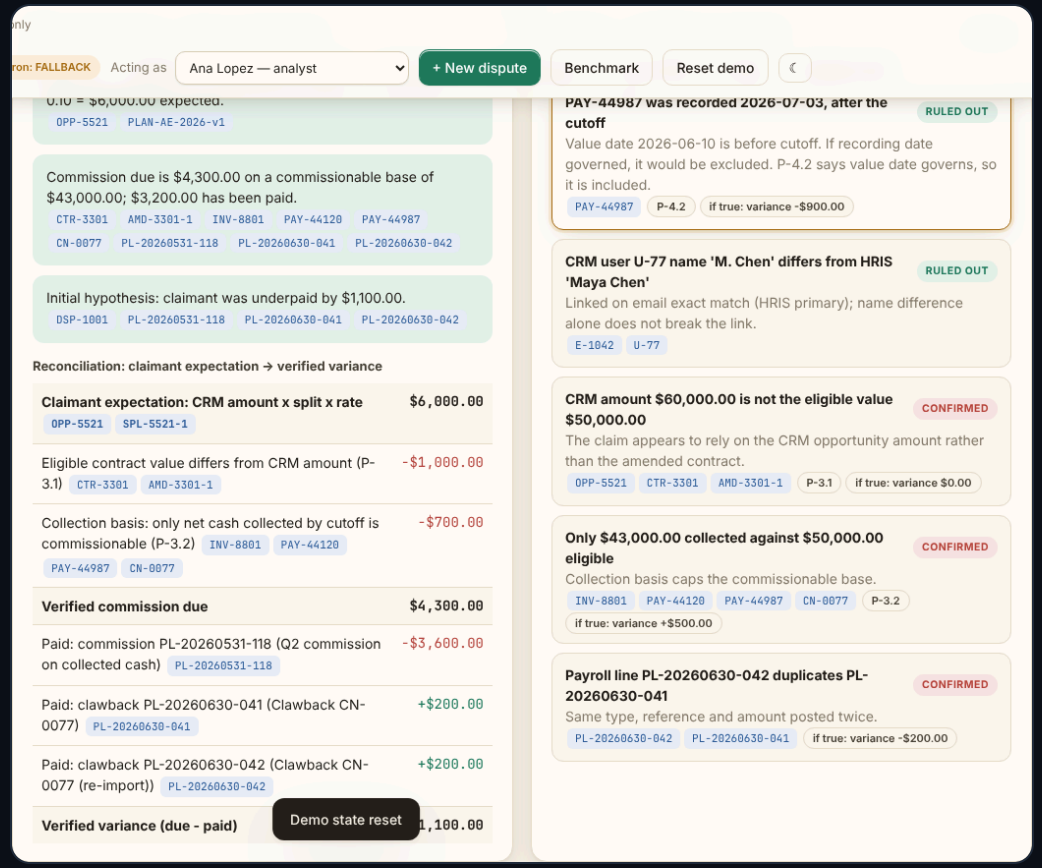
**Model**
NVIDIA Nemotron via OpenAI-compatible API, at most 2 calls per run, cached

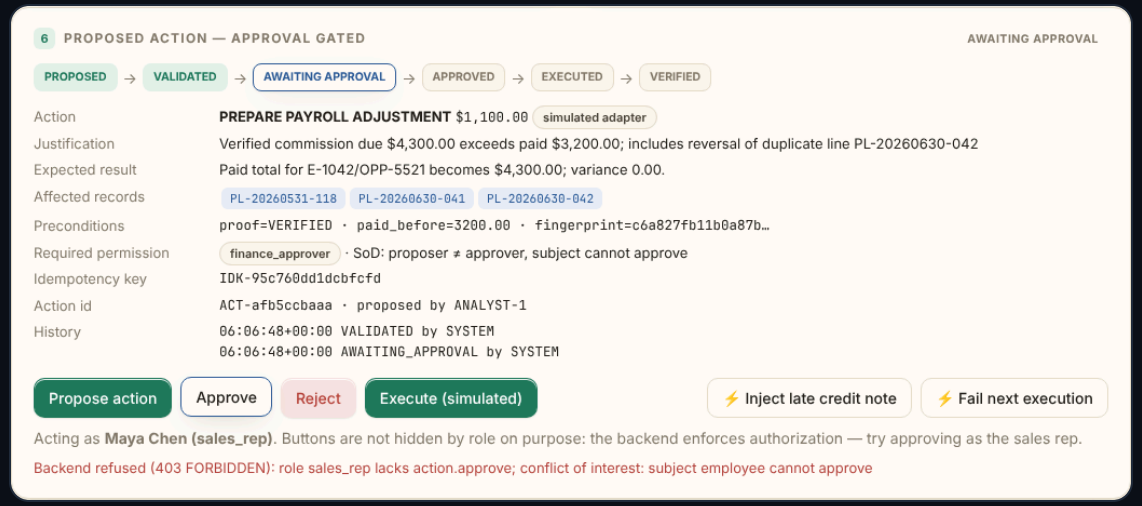
**Evidence**
JSONL traces in `logs/`, 3 samples, 76 tests, a 30-case benchmark, and **dispute intake** so any new case runs the full pipeline

**Run**
`python3 run.py` or Docker; `run.py --check-integrations`

PRODUCT VISUALS

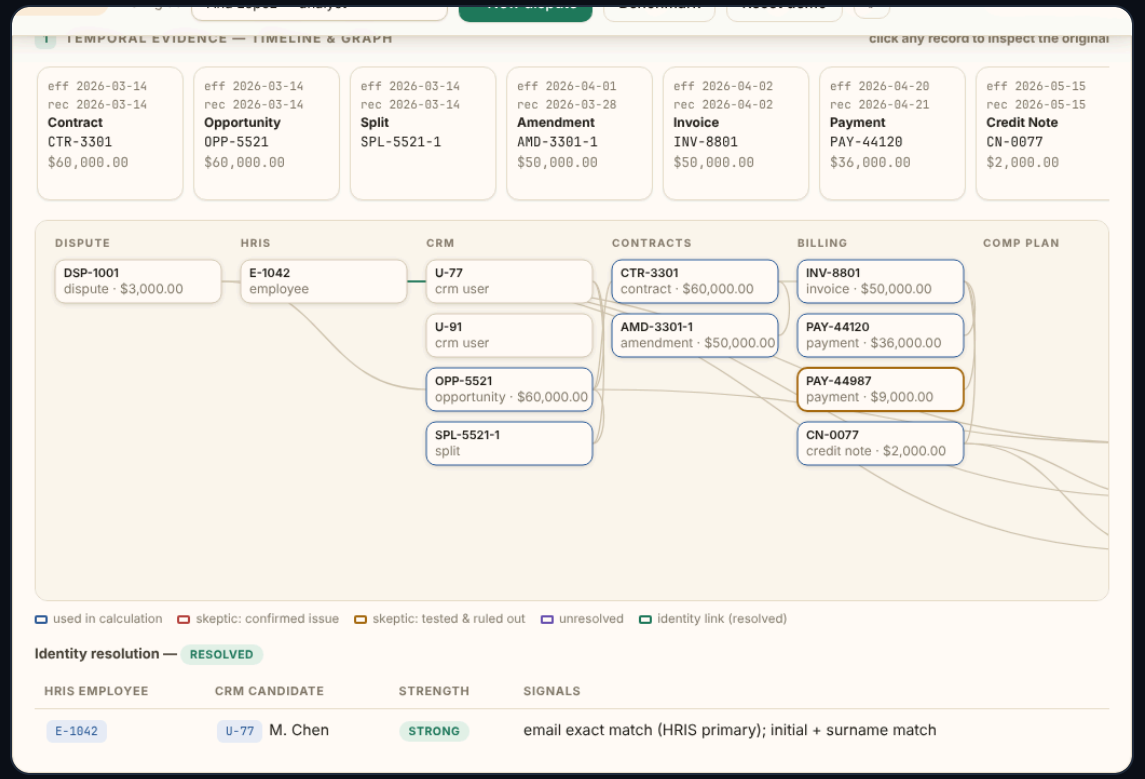
# The Decision Room

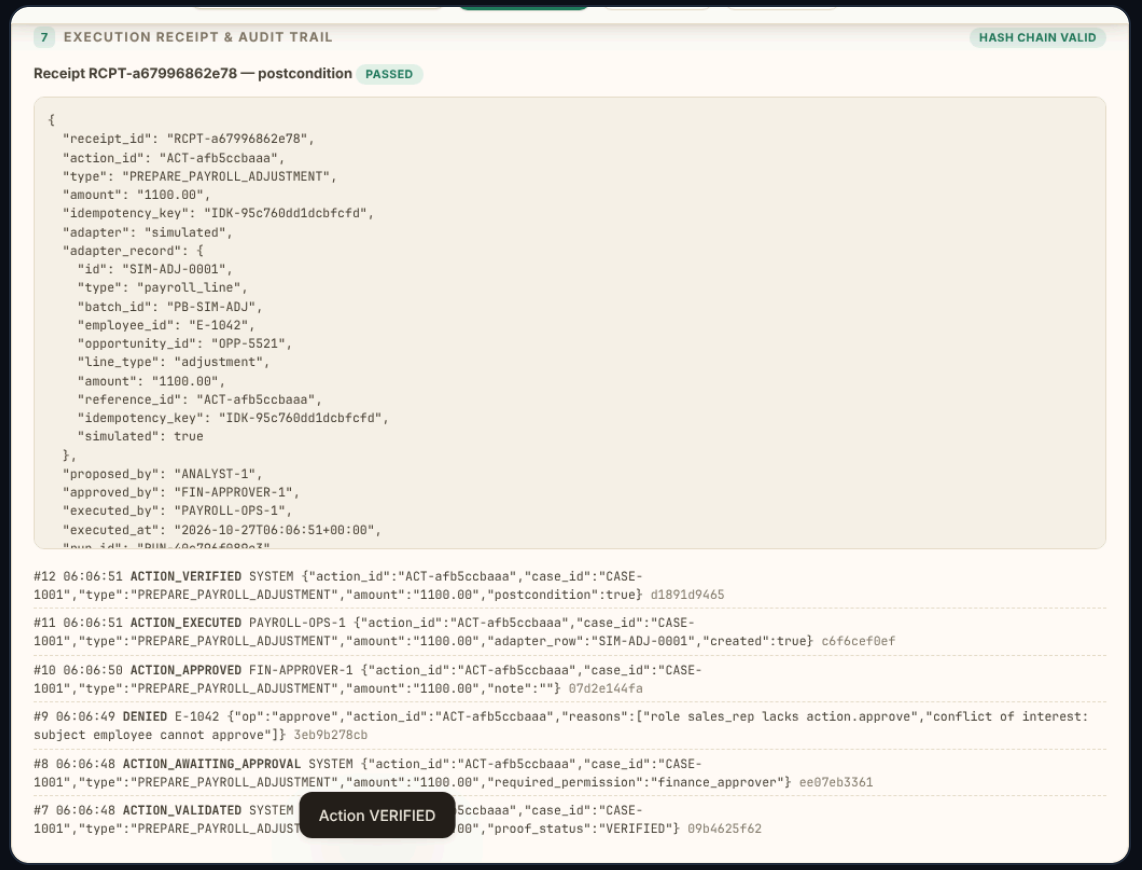


PRODUCT VISUALS

# Evidence graph and execution receipt



FUTURE ENHANCEMENTS

# Next

- A live Zetaris semantic layer over real HRIS / CRM / billing / payroll connectors
- Precedent ranking tuned on reviewer feedback through H-MEM's trust ledger, and a learned injection classifier alongside Scout's patterns
- A durable event store, SSO with role mapping, and real payroll adapters behind the same gate
- Multi-plan support: accelerators, quotas, multi-period clawbacks
- An independent, externally labelled benchmark set

TRY IT

# python3 run.py

No dependencies and no credentials. It opens on CASE-1001 at http://127.0.0.1:8360.

github.com/victrvondoom/Open-Agent

python3 -m unittest discover -s tests · python3 -m payproof.bench